# Jakhangir Tynshimov

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## Education

**Oregon State University** **Expected Jun 2028**
*B.S. Computer Science, Physics minor* *Corvallis, OR*
**Leadership:** President, Google Developer Group on Campus; Community Outreach Officer, ACM

## Experience

**Software Engineer Intern** Aug 2026 – Sep 2026
*Zyp, Inc. (www.zyp.co)* *Remote, Philippines*

- Built and launched Zyp's cross-border payments marketing site as a static **Astro 7 + TypeScript** site on Vercel: first commit to production in 4 weeks, authoring 67 of the 68 pull requests.
- Engineered a **GSAP ScrollTrigger**-scrubbed **Canvas 2D** video hero and a Canvas 2D globe with payment arcs (Natural Earth land built with d3-geo; 1.3 ms average draw), all without WebGL.
- Capped peak decoded-frame memory at 149 MB (from 995 MB) with a sliding **ImageBitmap** decode window; diagnosed scroll stutter as decode starvation by trace profiling, then a direction-aware, byte-budgeted window and WebP scrub path cut medium-wheel fallback draws from 69–83% to 0–1% (headless benchmark).
- Reached **Lighthouse** 100/100/100/100 desktop, 98/100/100/100 mobile (preview build; LCP 0.7 s / 2.2 s, CLS 0); built a **Playwright** + **axe-core** e2e suite for Chromium, WebKit and mobile: 128 tests per project, 0 axe violations.
- Directed implementation through written stage briefs executed by AI coding agents (**Claude Code**, **Codex**, **Gemini**); reviewed diffs against 13 written architecture invariants and a real-Safari gate before merge.

**Full-Stack & Hardware Developer** Nov 2024 – Sep 2025
*Automated Monitoring Solutions (case study)* *Aktau, Kazakhstan*

- Engineered a full-stack pre-shift inspection and workforce-verification platform (identity check, temperature, breath alcohol, SpO2) with an admin dashboard for attendance and health logs, using **Python, React, and TypeScript**.
- Assembled, wired and **soldered** the **Raspberry Pi** edge units and installed camera and sensor units on-site.
- Deployed the platform on **AWS** with **Docker** and **Portainer**; integrated **YOLO** and **Frigate** for real-time human detection and developed the **API** endpoints connecting the edge devices to the central cloud dashboard.

**President, Rocket Club** (portfolio) | *C++, Arduino, Fusion, 3D printing* 2022 – 2024

- Led the club: **Fusion**-modelled airframe **3D-printed** in sections; **C++** avionics bench-tested before integration.

## Projects

**Haptic Grade 1 Braille reader for web pages** May 2026
*Raspberry Pi (gpiozero), Circuit Design, Python, Flask, Playwright, Gemini* | GitHub

- **2nd Place Overall of 71 submissions, QuackHacks 3** (team of 3). Designed and hand-**soldered** the motor driver board (8 motor channels, 5 debounced keys) for six vibration motors that play **Grade 1 Braille**.
- The team's **Playwright** + **Gemini** extractor turned a web page into a navigable tree, served by **Flask** on the Pi.

**AR sound-direction glasses for deaf and hard-of-hearing users** May 2026
*Raspberry Pi, Python, Flask, faster-whisper, Gemini, PWA* | GitHub

- **3rd Place, Google "Best Use of Gemini" track, BeaverHacks 2026** (Oregon State; team of 2). Brought up the **Raspberry Pi** with a **reSpeaker XVF3800** USB 4-mic array and built the **3D-printed** glasses for a wearable that ran **faster-whisper** on-device and sent bearing, volume and transcript to **Gemini Flash** for JSON triage.
- Built the phone **AR overlay**: a camera **PWA** polling **Flask** `/state` for bearing, volume and Gemini's 8-word triage.

**Daily NBA-picks mobile app with ML player projections** Devpost | Feb – Mar 2026
*React Native, Supabase, Python* | GitHub

- **Winner, GDG Winter 2026 Hackathon** (113 participants; team of 3). Authored 66 of 74 commits: built the **React Native** profile screen and ML player cards and wired the **ML player-projection API** into game details.

## Technical Skills

**Languages**: Python, C++, JavaScript, TypeScript, SQL
**Frameworks & Libraries**: Astro, React, React Native, Next.js, Node.js, Flask, GSAP, Playwright, Supabase
**AI & Vision**: OpenAI/Anthropic SDKs, YOLO, ONNX Runtime Web, OpenCV
**Tools & Platforms**: Docker, AWS, Google Cloud Platform, Git, Unix/Linux, Vercel, Raspberry Pi
**Hardware**: Circuit design, soldering, sensor integration (GPIO, USB), 3D printing, Cura, Fusion